

### [CSE110 Spring 2024 Lab Exam - 0X Tentative Solutions and Rubrics](#)

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

#### **[SET-A]**

#### **SOLUTION**

```
import java.util.Scanner;

public class DataUsageTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter N: ");
        int N = sc.nextInt();

        int dataScore = 0;

        for (int i = 1; i <= N; i++) {
            System.out.print("Enter data for day " + i + ": ");
            int data = sc.nextInt();

            if (data < 50 || data > 5000) {
                System.out.println("Invalid data amount, " + data + ".");
                i--;
                continue;
            }

            if (data % 4 == 0 || data % 10 == 5) {
                dataScore += data;
            } else {
                dataScore -= data;
            }
        }

        int scoreCopy = dataScore;
        if (scoreCopy < 0) scoreCopy *= -1;

        int evenCount = 0;
        while (scoreCopy > 0) {
            int digit = scoreCopy % 10;
            if (digit % 2 == 0) evenCount++;
            scoreCopy /= 10;
        }

        System.out.println("Final Data Score: " + dataScore);
        System.out.println("Number of even digits: " + evenCount);
    }
}
```

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**RUBRIC**

| SL | Points To Meet                          | Marks [10] |
|----|-----------------------------------------|------------|
| 1  | Scanner used properly, N input taken    | 1.5        |
| 2  | Loop runs N times with re-prompt        | 1          |
| 3  | Proper Range check (50–5000)            | 1.5        |
| 4  | Score updated using rules               | 2.5        |
| 5  | Even-digit counting + negative handling | 2.5        |
| 6  | Output displayed correctly              | 1          |

**[SET-B]**

**SOLUTION**

```
import java.util.Scanner;

public class WaterConsumptionTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter N: ");
        int N = sc.nextInt();

        int hydrationScore = 0;

        for (int i = 1; i <= N; i++) {
            System.out.print("Enter water for day " + i + ": ");
            int water = sc.nextInt();

            if (water < 100 || water > 10000) {
                System.out.println("Invalid water amount, " + water + ".");
                i--; // re-prompt this day
                continue;
            }

            if (water % 5 == 0 || water % 10 == 2) {
                hydrationScore += water;
            } else {
                hydrationScore -= water;
            }
        }

        int scoreCopy = hydrationScore;
        if (scoreCopy < 0) scoreCopy *= -1;

        int oddCount = 0;
        while (scoreCopy > 0) {
            int digit = scoreCopy % 10;
            if (digit % 2 != 0) oddCount++;
            scoreCopy /= 10;
        }

        System.out.println("Final Hydration Score: " + hydrationScore);
        System.out.println("Number of odd digits: " + oddCount);
    }
}
```

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**RUBRIC**

| <b>SL</b> | <b>Points To Meet</b>                                     | <b>Marks [10]</b> |
|-----------|-----------------------------------------------------------|-------------------|
| 1         | Scanner creation & input N taken                          | 1.5               |
| 2         | Loop runs N times, reprompt on invalid input              | 1                 |
| 3         | Range check 100–10,000 ml                                 | 1.5               |
| 4         | Score updated based on divisible by 5 or ends with 2 rule | 2.5               |
| 5         | Odd digit counting with negative handling                 | 2.5               |
| 6         | Final output formatting correct                           | 1                 |